

Control Estadístico de la Calidad

Examen Argumentativo

Nombre: _____ Matrícula: _____

Instrucciones: Muestra cómo obtienes tu respuesta. Puedes usar Excel o Minitab para tus cálculos de probabilidades o percentiles, pero no uses otros archivos o hagas uso de la IA. Respeta el código de ética del curso y del Tecnológico de Monterrey.

1. A Shewhart $\bar{X}$ chart is used with subgroup size $n = 5$. The process has $\bar{\bar{X}} = 67$ and $\bar{R} = 7$. Calculate the upper control limit (UCL) of the chart.
2. A process has specification limits $LSL = 85$ and $USL = 105$. The estimated process standard deviation is $\sigma = 2$. Calculate the potential capability index C_p .
3. A Shewhart control chart has center line 106, $UCL = 120$ and $LCL = 92$. A new sample statistic is 124.
What is the best conclusion?
 - (a) The process variability has decreased.
 - (b) No action is needed; the point is within limits.
 - (c) The process is centered and capable.
 - (d) The process is out of statistical control.
4. In an inspection of 200 units, 27 defective units were found. The fraction defective is $p = 0.135$. Which interpretation is correct?
 - (a) The process is automatically in statistical control.
 - (b) The process capability index equals 0.135.
 - (c) Approximately 13.5% of the inspected units were defective.
 - (d) Exactly 86.5% of future units will be conforming.
5. A process has $C_p = 1.8$ and $C_{pk} = 1.2$. Which conclusion is most appropriate?
 - (a) The process is out of statistical control.
 - (b) The specification limits are wider than the process spread because $C_{pk} > C_p$.
 - (c) The process spread is adequate, but the process is not perfectly centered.
 - (d) The process is perfectly centered between specifications.